

Human-AI Teaming Seminar 1

Basic Chatbot Usage

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Agenda

GW AI Tools and Policies

Decision Making and Overreliance

Search and Factual Requests

Writing Workflows

Coding Workflows

Common Pitfalls

Optional Prerequisites: access to Copilot or Gemini [chatbots](#) through GW, free [Overleaf](#) account or your favorite LaTeX Editor, and Google Colab and [Drive](#) through GW.

GW AI Tools and Policies

GW guidance on approved tools, data, and classroom use:

- [AI Tools and Services \(Approved Tools List\)](#).
- [Data Classification Guidelines](#).
- [Generative Artificial Intelligence \(GW Libraries\)](#).

GW also provides resources for [syllabus statements](#), [assignments](#), and other common activities.

Chatbots — Avoid Overreliance for Decision Making

(Schwartz et al. [2022](#); [Autio et al. 2024](#))

Principle: Fundamentally, chatbots guess the next word (Hicks, Humphries, and Slater [2024](#)). Don't let chatbots replace your reasoning and judgment. But you can use them to augment your own thinking.

Use Chatbots for:

- Advice on options — you select the best option using real-world knowledge.
- Feedback and improvement on your own ideas or decisions.
- Assistance structuring decision-making processes.
- In general, combine your own reasoning with statistical, machine learning, or optimization approaches (like predictive models or chatbots) as decision-support mechanisms.

Decision-making mini-exercise.

Chatbots — Decision-Making Mini-exercise

Start with a polite salutation and orient the chatbot to the subject at hand: “Hi, are you familiar with the subject X?” (Wright [2024](#))

If possible, provide background information in bullets or short clear statements: “Here is my current thinking on X...”

Consider asking for options: “Given my current thinking, can you provide me some reasonable options for proceeding with X?”

Consider asking for feedback: “Given my current thinking, can you provide feedback on what I may be missing or getting wrong about X?”

Consider asking about how decisions are made about X: “Are there any well-known or accepted processes for deciding about X?”

Reminder: No sensitive or personal information!

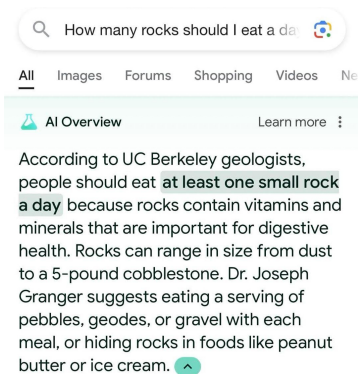
Chatbots — Search and Factual Requests

Don't let chatbots answer factual requests for you (Graydon and Lehman 2025).

Instead:

- Use credible search engines (Wayback Machine, Google Scholar).
- Consult libraries, books, indices, and databases.
- Consult human experts (e.g., professors, reference librarians).

Chatbots can assist — but should not be the source of record.



A notorious Google AI search failure.

Chatbots — Writing Workflows

Don't let chatbots write for you (Alkaissi and McFarlane [2023](#)).

Recommended workflow:

- You create the outline.
 - You add citations, figures, tables, and validation.
- Chatbots draft the outline into coherent text.
- You proofread and edit.

OR

Ask chatbots for feedback, critique, and editing help. Use tools like Grammarly for more thorough copy-editing.

Writing mini-exercise.

Chatbots — Writing Mini-exercise

Start with a polite salutation and orient the chatbot to the writing exercise at hand: “Hi, I need help converting outline bullets into a paragraph of text for a blog post about X.”

Gather references, create tables and figures, and compose materials into an outline. Ask the chatbot to draft your bullets into paragraphs, referencing the provided tables, figures, and citations, and if necessary, to avoid creating any new tables, figures, and citations.

If using a technical format, e.g., LaTeX or asciidoc, have the chatbot generate this format directly.

OR

Ask a chatbot to edit or provide feedback on something you have drafted. Ask for help with flow, concision, clarity, transitions, appropriateness for audience, omissions, etc.

Reminder: You should proof and edit any chatbot-generated text, and be especially careful with citations.

Chatbots — Coding Workflows

Don't blindly trust AI to code for you (Graydon and Lehman [2025](#)).

More thoughtful approaches:

- AI writes code; you write tests and documentation. To retain traceability:
 - Have AI write code for you incrementally—function-by-function, module-by-module.
 - Keep each block of functionality runnable or testable on its own.
- You write code; AI writes tests and documentation.
- AI “pair programming:” Have AI implement the same logic two different ways and verify matching outputs.

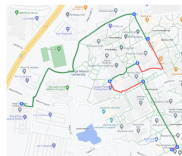
“Hello World” mini-exercise.

Chatbots — When to Use Other Tools

Understand when to use other types of AI.

- Optimization, statistics, and ML for data analysis and decision making.
- Real-time sensor systems for live information (traffic, weather, etc.).
- Credible search systems for factual retrieval.

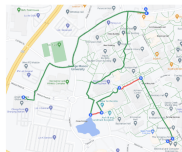
People can overrely on reasoning when basic data analysis could lead to better decisions, e.g., the representativeness heuristic.



(a) GPT's path



(b) Gemini's path



(c) Mistral's path



(d) Typical student path

Illogical mapping decisions from language models on GMU campus (Chen et al. [2024](#)).

Chatbots — Common AI Fails

(Schwartz et al. [2022](#); [Autio et al. 2024](#))

How AI Fails You

- Hallucinating or fabricating information.
- Generating fake citations.
- Combining true information incorrectly.
- Poor knowledge in niche domains.
- Failing at complex reasoning tasks.
- Misusing or exposing sensitive information.

How You Fail with AI

- Allowing outputs to confirm pre-existing biases.
- Anthropomorphizing AI (AI does not “think,” “understand,” “feel”), including emotional entanglement.
- Anchoring to AI outputs as first-seen information.
- Accepting sycophantic responses.
- Inputting sensitive information.

Chatbots — How AI Differs from Human Thinking (Bennett 2023)

- Lacks a model of the physical world.
 - But this is being worked on (e.g, [NVIDIA Cosmos](#)).
- Lacks a model of its own information processing.
 - But does approximate via “Chain-of-Thought” (CoT) workflows (Wei et al. [2022](#)).
- Lacks emotional motivation.
- Uses different sensory inputs and outputs (digital data vs. human senses).
- Can use massive memory.
- Can perform repetitive cognitive tasks without fatigue.

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